INCH-POUND
A-A-50455
26 February 1992

COMMERCIAL ITEM DESCRIPTION

PORTABLE FILTRATION CART

The General Services Administration has authorized the use of this commercial item description.

1. ABSTRACT

1.1 This commercial item description covers the requirements for 10, 16, and 25 gallons per minute (gpm) rated flow capacity filtration carts used in transferring petroleum base, water-glycol, and triaryl phosphate ester hydraulic fluids into ship hydraulic systems as well as cleaning the ships hydraulic systems that are contaminated with particulate, water, or both. It shall recirculate fluid in a contaminated system until cleanliness levels are attained which provide satisfactory component performance.

2. SALIENT CHARACTERISTICS

2.1 Physical.

Max. weight:
(a) 140 pounds (dry) for 10 gpm flow cart.
(b) 155 pounds (dry) for 16 gpm flow cart.
(c) 170 pounds (dry) for 25 gpm flow cart.

Max. dimensions: Height: 48 inches.
Width and Length: Width and length dimensions of the cart shall be such that it passes through a 24 inch diameter hatch.

2.2 Fabrication.

(a) All components shall be rigidly secured to a frame. The frame shall be equipped with rubber-tired wheels, 10 inches in diameter, and be designed for mobility and handling ease.
(b) The electric motor, coupling, and pump shall be accurately aligned and designed to accommodate the residual misalignment found under typical service conditions.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Commander, Naval Sea Systems Command, Attn: 55Z33, Washington, DC 20362-5101.

AMSC N/A FSC 4330

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(c) The unit shall operate in either the vert ical plane or the horizontal plane.

(d) The unit shall be constructed of the materials that are compatible with fluids in accordance with MIL-H-17672, MIL-H-19457 and MIL-H-22072.

(e) The unit shall be equipped with a drip pan to collect oil dripping during and after unit operation, and to keep the working area safe and clean.

2.3 Electrical.

(a) Electrical enclosures shall meet or exceed NEMA MG1 standards.

(b) Electrical power requirements for the cart rated flow shall be as follows:

TABLE I. Power requirement.

Rated flow (gal/min)	Motor hp	Electrical power
10	0.75	115 volts, 60 Hz 16 amp draw (max) under full load; single phase
16	1.50	
25	3.00	440 volts, 60 Hz 10 amp draw (max) under full load; triple phase

(c) Electrical motor shall be a totally enclosed fan cooled (TEFC) type; continuous duty.

(d) An oil-tight, permanently mounted, on/off switch shall be provided.

(e) A cord, 90 inches minimum length, NEMA type 50 (heavy duty, oil-resistant) with a ground cable capable of conducting the full load motor current shall be provided. One end of the cord shall be permanently attached to the filter unit. The other end shall be fitted with a NEMA type, three-prong, dead front plug.

(f) The electric motor shall automatically stop when the filter bypass valve starts to crack open, that is, when the dirt alarm indicator is at the junction of the yellow and red bands. A manual reset shall also be provided to prevent the motor from automatically restarting after it stops. The reset shall meet the applicable standards specified for the other electrical components above.

2.4 Oil delivery.

(a) An integral part of the unit shall be a positive displacement pump capable of delivering rated flow of oil at 215 centistoke (cSt) viscosity and 100 pounds per square inch (lb/in^2) minimum pump discharge pressure continuously. The rated flow of unit shall be 10, 16, or 25 gallons per minute (gal/min) as specified by the contracting activity

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(b) The pump shall be self-priming and shall operate with a vacuum of up to 15 inches of mercury at the inlet.
(c) The portable filter cart shall be provided with a safety shut-off to prevent pump pressure in excess of that which can be safely contained by the unit's individual components.

2.5 Filtration.

(a) Two primary pressure filters connected in series, with a minimum burst rating of 1000 lb/in^2 gauge, shall be integral parts of the unit to provide dual filtration for both particulate and water contaminant removal.
(b) The filter bowls shall be attached to the filter heads in a manner to permit removal and cleaning of the bowls and servicing of the elements without breaking pipelines, The servicing requirements of the filters shall be simple and shall not require the use of special tools for removing, cleaning, replacing elements, and remounting filter bowls.
(c) The bypass valve shall be fast-acting, single stage, and of high capacity design to protect the filter elements from excessive differential pressure under all transient and steady flows found in service.
(d) The bypass valve shall have a cracking pressure of 35 to 40 lb/in^2 gauge and a full flow pressure drop of 60 lb/in^2 maximum at rated flow, 215 centistoke (cSt) viscosity fluid.
(e) Maximum leakage of the bypass valve at 20 lb/in^2 differential pressure shall not exceed 60 milliliters per minute with 43 cSt oil.
(f) The annular clearance between the inside diameter of the housings and the outside diameter of the elements shall be not less than 3/16 inches.
(g) The fluid flow through the filter elements shall be from the outside to the inside.
(h) Filter elements:

(1) For particulate removal: Each filter housing shall accept two filter elements that are in accordance with MIL-F-24702 and MIL-F-24702/1.
(2) For water removal: Each filter housing shall accept two filter elements of any type. Water removal elements must conform to the dimensions specified in MIL-F-24702/1.
(3) The two elements installed in each housing shall be connected by the use of reusable element-connector that is in accordance with MIL-F-24702/1.

2.6 Differential Pressure indicator (dirt alarm).

(a) The filters shall be provided with a clearly visible mechanical differential pressure indicator. The indicator shall actuate either by direct mechanical linkage to the bypass valve or by the differential pressure across the filter elements.
(b) The indicator shall reflect dirt buildup (pressure differential) in the elements and shall register against a three-color scale

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(c) The three colors shall read:

(1) GREEN	OK (indicating full flow filtration).
(2) YELLOW	CHANGE (indicating full flow filtration, element approaching by-pass cordition) .
(3) RED	BYPASS (indicating oil bypassing the filter elements).

2.7 Unit ports. The unit shall be equipped with 1-1/4 inch interchangeable type steel quick disconnects in accordance with NFP(A) B93.113M. The companion mating parts of the quick-disconnect couplings shall be provided with each unit. The inlet fitting shall be male and outlet fitting shall be female.

2.8 Gauges. Two gauges meeting or exceeding the requirements of ANSI B40.1, grade B, 2 inches in diameter shall be provided with the unit. Both shall be securely mounted to the unit and shall be visible to the operator while the unit is in use. The gauge line to each gauge shall be equipped with a gauge isolation/ vent valve. The gauges shall be connected to the unit piping with SAE straight-thread fitting as follows:

(a) One gauge shall have a range of O to 200 lb/in^2 and shall be connected to the pump discharge piping or filter inlet so that pump discharge pressure can be monitored.
(b) The other gauge shall have a range of O to 30 inches of mercury and shall be connected to the pump inlet piping or the suction filter outlet connection so that pump suction (vacuum) condition can be monitored.

2.9 Oil sampling valves. Two vent and test valves, in accordance with MIL-V-24695 and MIL-V-24695/1 shall be provided in the unit piping or filter base to allow oil samples to be taken. One valve shall be located on the inlet side of the filter and the other on the outlet side.

2.10 Seals. O-ring seals shall be fluorocarbon rubber and shall conform to an SAE or military specification with part numbers of standard sizes in accordance with SAE AS 568.

2.11 Fasteners. Screws and bolts shall be selected in accordance with MIL-STD-1251. Other fasteners shall be selected in accordance with MIL-STD-1754. Washers shall be selected in accordance with MIL-STD-1764. Screws, bolts, washers, nuts, and other fasteners in accordance with ANSI, ASTM and SAE part standards with part identification numbers may be used when the part identification numbers identify the size, material, coating (if any) and all other characteristics (including test and inspection) necessary to procure the fasteners.

2.12 Protective devices:

(a) Pump shall be protected with a 100 mesh inline strainer or separator that can be serviced without breaking lines and draining fluid from the unit.
(b) Guards shall be provided for powered moving parts to prevent contact by operators.

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(c) The unit shall be equipped with an electric motor overload protection feature which will automatically stop the motor under an overload condition.

2.13 Identification

(a) The direction of flow through the filter shall be clearly marked on filter housing.

(b) The following acquisition information shall be permanently identified on the unit:

(1) CID number.
(2) Manufacturer's name.
(3) Part number.
(4) CAGE number.

(c) Appropriate warning plates shall be conspicuously located where hazards exist. A warning plate providing caution against the use of water removal filter elements in water-glycol systems shall be provided as follows:

"WARNING: DO NOT USE WATER REMOVAL FILTER ELEMENTS ON WATER-GLYCOL SYSTEMS"

(d) Instruction plates shall be installed showing the assembly of the filter elements, using the element connectors, within the filter bowl ,

2.14 Identification of standard parts. O-ring seals shall be identified either by military standard part numbers or by SAE standard part numbers which reflect the required material and limit sizes to standard sizes. All fasteners shall be identified by a military standard part number or an industry part number which identifies both size and material.

2.15 Paint. The unit shall be painted with an epoxy paint.

2.16 Manuals. Standard commercial manuals including operating, service, maintenance instruction, and a parts list shall be furnished with each unit.

3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.2 First article. First article testing shall be performed by the unit manufacturer to demonstrate compliance with this specification. Supporting data shall be provided for the following tests:

(a) oil delivert, with approved filter elements (see 2.5(h)), shall be as specified in 2.4. For record purposes, identify the pump discharge pressure at the full flow test condition.
(b) Bypass valve cracking pressure and full flow pressure drop shall be as specified in 2.5(d).

4.3 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics of this commercial item description, and the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING

5.1 Preservation. Packing. labeling. and marking. Presentation, packing, labeling, and marking shall be as specified in the contractor order.

6. NOTES

6.1 Ordering data. Purchaser must specify the following:

(a) Title, number, and date of this commercial item description.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 6.4.1).
(c) Flow rate required (see 2.4).
(d) Electrical power required (see 2.3(b)).

6.2 CID based Part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. The part number consists of the basic number of this commercial item description plus applicable flow rate and power requirement numbers selected from table I.

EXAMPLE: A-A-50455-16-115 is a Filtration Cart with a 16 gal/rein flow rate and 115 volts power requirement.

6.3 Acceptable units. The brand names and manufacturer's part numbers of filtration carts which are known to be acceptable and of current manufacture are available from: Commander, Naval Sea Systems Command, Attn: 56W1, Washington, DC 20362-5101. Manufacturers desiring listing of their products should submit drawings and test data that demonstrate compliance with this specification.

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6.4 Sources of documents

6.4.1 Government documents.

SPECIFICATIONS

MILITARY

MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited.
MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic.
MIL-H-22072 - Hydraulic Fluid, Catapult, Nato Code Number H-579.
MIL-V-24695 - Valve, Hose Assembly and Adopter, Vent and Test Hydraulic Service, General Specification For.
MIL-V-24695/1 - Valve, Vent and Test, Hydraulic Service.
MIL-F-24702 - Filter Elements, Hydraulic, Disposable, General Specification for. (Metric)
MIL-F-24702/1 - Filter Elements, Hydraulic, Disposable. (Metric)

STANDARDS

MILITARY

MIL-STD-1251 - Screws and Bolts Preferred for Design, Listing of.
MIL-STD-1754 - Fastening Devices Preferred for Design, Listing of.
MIL-STD-1764 - Washers Preferred for Design, Listing of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

6.4.2 Non-Government documents.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B40.1 - Gauges-Pressure Indicating Dial Type - Elastic Element. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W. , Suite 300, Washington, DC 20037.)

NATIONAL FLUID POWER ASSOCIATION (NFP(A))

B93.113M - Hydraulic Fluid Power-Quick-Action Couplings - Dimensions and Requirements.

(Application for copies should be addressed to the National Fluid Power Association 333 N Mayfair Road, Milwaukee, WI 53222.)

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SO CI ETY OR AUTOMOTIVE ENGINEERS (SAE)
AS 568 - Aerospace Size Standard for G-Rings. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

Preparing activity:
Navy - SH
(Project 4330-N087)

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PORTABLE FILTRATION CART

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
TECHNICAL POINT OF CONTACT: Mr. Ratee Patel SEA 56W16	(1) Commercial 703-602-1595	(2) AUTOVON 332-1595
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 55Z3 Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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